

Full Solution Packet

Multiple real points at infinity force non-flat moments

Status: candidate full solution, likely valid

Target: the conjecture after Example 7.2.3 of arXiv:2407.06017

Source and exact conjecture

The source is Baldi–Blekherman–Sinn, *Nonnegative Polynomials and Moment Problems on Algebraic Curves* [1]. Under its standing assumptions, the projective closure $\overline{X} \subset \mathbb{P}^2$ is a smooth totally real plane cubic and $X = \overline{X} \setminus S$ is an affine part, where $S = \overline{X}(\mathbb{R}) \setminus X(\mathbb{R})$ is the finite set of real points at infinity.

We now construct an example which illustrates that $\overline{X}(\mathbb{R})$ having a unique point at infinity is a necessary assumption in Theorem 7.2.2.

Example 7.2.3. Let $\widetilde{X}(\mathbb{R}) \subset \mathbb{R}^2$ be the real locus of the affine totally real plane cubic, and assume that $\widetilde{X}(\mathbb{R})$ is connected. Let $L \in \text{int } \mathcal{M}_{2d}(\widetilde{X}(\mathbb{R})) \subset \mathbb{P}_{\widetilde{X}, \leq 2d}^\vee$.

We deduce from Lemma 6.2.3 that L admits exactly two distinct representations

$$L = \sum_{i=1}^{3d} a_i \text{ev}_{\widetilde{A}_i} = \sum_{i=1}^{3d} b_i \text{ev}_{\widetilde{B}_i}$$

using $3d$ points form the projective closure of $\widetilde{X}(\mathbb{R})$. Now consider the (real) line between A_1 and B_1 , and apply a projective linear change of coordinates in such a way the line between $\widetilde{A}_1$ and $\widetilde{B}_1$ is sent to the line at infinity. Let $X(\mathbb{R}) \subset \mathbb{R}^2$ be the affine real locus of $\widetilde{X}(\mathbb{R})$ under this change of coordinates.

We have $L \in \text{int } \mathcal{M}_{2d}(X(\mathbb{R})) = \text{int } \mathcal{M}_{2d}(\widetilde{X}(\mathbb{R}))$, and as before L admits only two representations

$$L = \sum_{i=1}^{3d} a_i \text{ev}_{A_i} = \sum_{i=1}^{3d} b_i \text{ev}_{B_i} \in \mathbb{P}_{\widetilde{X}, \leq 2d}^\vee = \mathbb{P}_{X, \leq 2d}^\vee$$

where A_i, B_i denote the images of $\widetilde{A}_i, \widetilde{B}_i$ after the change of coordinates above. Notice that both representations use a point at infinity, i.e. A_1 and B_1 respectively, by construction.

This shows that $L \in \text{int } \mathcal{M}_{2d}(X(\mathbb{R}))$ cannot be represented using $3d$ evaluations at points of the affine curve $X(\mathbb{R})$, and thus L does not admit a flat extension. Therefore, we cannot remove the condition that $\overline{X}(\mathbb{R})$ has a unique point at infinity from Theorem 7.2.2, to conclude that the flat extension condition is sufficient. Notice also that the above discussion implies that $\mathcal{C}_{X, \leq 2d}(L) = 3d + 1$, but $\mathcal{C}_{\widetilde{X}, 2d} = 3d$, and thus $\mathcal{C}_{\widetilde{X}, 2d} < \mathcal{C}_{X, \leq 2d}$. It is therefore not possible to compute the affine Carathéodory numbers using solely the projective information.

We conjecture that such pathological behavior happens for all affine plane cubics with at least two real points at infinity.

For degree $2d$, put $r = 3d$. The source constructs one connected example with a moment functional whose projective Carathéodory number is r but whose affine Carathéodory number is $r + 1$, and conjectures that this pathology occurs for every affine plane cubic with at least two real points at infinity. Equivalently, ordinary positive flat extension is not sufficient in all such cases.

Result

Theorem 1. Let $\overline{X} \subset \mathbb{P}^2$ be a smooth totally real plane cubic, let $X = \overline{X} \setminus S$, and suppose that S contains at least two distinct real points. For every integer $d \geq 1$, there exists

$$L \in \mathcal{M}_{2d}(X(\mathbb{R}))$$

which has no positive flat extension from degree $2d$ to degree $2d + 2$.

If $\overline{X}(\mathbb{R})$ is connected, L may be chosen so that

$$\mathcal{C}_{\overline{X}, 2d}(L) = 3d, \quad \mathcal{C}_{X, \leq 2d}(L) = 3d + 1.$$

Together with the disconnected case proved in Theorem 7.2.4 of the source, this proves the conjecture after Example 7.2.3.

Proof intuition

Choose two prescribed points $P, Q \in S$. On the circle $\overline{X}(\mathbb{R})$, construct a reduced real divisor

$$D = T_1 + \cdots + T_{6d} \in |\mathcal{O}_{\overline{X}}(2d)|$$

which contains P, Q in positions of opposite parity. If q is a real section with divisor D , the residue theorem applied to $(h/q)\omega$ gives the unique relation among the evaluations at the T_i . Its coefficients alternate in sign around the circle. Hence the relation splits into two positive $3d$ -atomic representations of one functional L , one containing P and the other containing Q .

A generic choice of D makes L an interior projective moment functional. The source's Lemma 6.2.3 then says that these are its only two minimal projective representations. Thus no $3d$ -atomic representation avoids S . On the other hand, the source's Lemma 6.2.6 supplies a $(3d+1)$ -atomic representation avoiding any prescribed finite set, hence avoiding S . This proves the claimed affine/projective rank gap and rules out a flat extension.

Two lemmas

Set

$$N = 6d, \quad r = N/2 = 3d, \quad \mathcal{L} = \mathcal{O}_{\overline{X}}(2d),$$

and write H for a hyperplane divisor on $\overline{X}$. Thus $\mathcal{L} \cong \mathcal{O}_{\overline{X}}(2dH)$, $\deg \mathcal{L} = N$, and $\dim H^0(\overline{X}, \mathcal{L}) = N$.

Lemma 1 (A prescribed alternating real divisor). *Assume that $\overline{X}(\mathbb{R})$ is connected. Given distinct $P, Q \in \overline{X}(\mathbb{R})$, there is a reduced totally real divisor*

$$D = T_1 + \cdots + T_N \in |\mathcal{L}|$$

containing P and Q , such that P and Q have opposite parity in the cyclic order of the T_i . Moreover, if

$$A = T_1 + T_3 + \cdots + T_{N-1}, \quad B = T_2 + T_4 + \cdots + T_N,$$

then the divisor can be chosen so that

$$2A \not\sim \mathcal{L}. \tag{1}$$

Proof. The connected real elliptic curve $\overline{X}(\mathbb{R})$ is a circle group. Choose a group coordinate $\theta : \overline{X}(\mathbb{R}) \rightarrow \mathbb{R}/\mathbb{Z}$. By Abel's theorem, a real divisor of degree N belongs to $|\mathcal{L}|$ precisely when the sum of its points has the fixed value corresponding to $\mathcal{L}$ in $\text{Pic}^N(\overline{X})(\mathbb{R})$.

Of the two open arcs between P and Q , choose one, denoted I , whose length in the group coordinate is at least $1/2$. Put all the remaining $m = N - 2 \geq 4$ points in I . On the ordered configuration simplex

$$a < t_1 < \cdots < t_m < b$$

for a lift $I = (a, b)$, the ordinary sum $t_1 + \cdots + t_m$ ranges through the interval (ma, mb) . Its length is $m(b - a) \geq 2$, so its image modulo $\mathbb{Z}$ is the whole circle. We can therefore choose distinct points in I so that, together with P, Q , their Abel sum is the class of $\mathcal{L}$. This gives $D \in |\mathcal{L}|$.

The number of chosen points on one arc from P to Q is either 0 or m , and m is even. Thus P, Q have opposite parity in the cyclic ordering of D .

It remains to impose (1). The fixed-sum configuration has positive dimension. Move one point assigned to A by a small amount t and an adjacent point assigned to B by $-t$. The Abel sum of D stays fixed while the Abel sum of A varies. The forbidden condition $2A \sim \mathcal{L}$ asks that $A - dH$ be one of finitely many 2-torsion classes. A generic small variation therefore avoids it while preserving distinctness and cyclic order. $\square$

Lemma 2 (Alternating residue circuit). *Let $D = T_1 + \cdots + T_N \in |\mathcal{L}|$ be reduced and totally real, written in cyclic order. There is a unique relation, up to nonzero scale,*

$$\sum_{i=1}^N c_i \operatorname{ev}_{T_i} = 0 \quad \text{on } H^0(\overline{X}, \mathcal{L}),$$

and every c_i is nonzero. After reversing the overall sign if needed,

$$c_1 > 0, \quad c_2 < 0, \quad c_3 > 0, \dots, c_N < 0.$$

Proof. Choose a real section $q \in H^0(\overline{X}, \mathcal{L})$ with $\operatorname{div}(q) = D$ and a nonzero real holomorphic differential ω on $\overline{X}$. For every $h \in H^0(\overline{X}, \mathcal{L})$, the meromorphic differential

$$\frac{h}{q} \omega$$

has simple poles precisely at the T_i . The residue theorem gives

$$0 = \sum_{i=1}^N \operatorname{res}_{T_i} \left(\frac{h}{q} \omega \right) = \sum_{i=1}^N c_i h(T_i).$$

All c_i are real and nonzero.

The line bundle $\mathcal{L} = \mathcal{O}_{\overline{X}}(d)^{\otimes 2}$ is a square, so its real restriction to the circle $\overline{X}(\mathbb{R})$ is trivial. In a real trivialization, q is a real function with simple zeros. Its sign changes at every T_i , and the signs of its derivatives at successive zeros alternate. Orienting the circle by the nowhere-zero form ω , the residue coefficient is a positive normalization of $1/q'(T_i)$. Hence the signs of the c_i alternate.

Finally, the evaluation map

$$H^0(\overline{X}, \mathcal{L}) \longrightarrow \bigoplus_{i=1}^N \mathcal{L}|_{T_i}$$

has kernel $H^0(\overline{X}, \mathcal{L}(-D)) = H^0(\overline{X}, \mathcal{O})$, which is one-dimensional. Both its domain and target have dimension N , so the cokernel, and therefore the displayed evaluation relation, is one-dimensional. $\square$

Proof of Theorem 1

Proof. The source's Theorem 7.2.4 already proves the required non-flat behavior when $\overline{X}(\mathbb{R})$ has two connected components. Assume from now on that $\overline{X}(\mathbb{R})$ is connected.

Choose distinct $P, Q \in S$ and apply Lemma 1. Relabel the cyclic order so that the residue signs in Lemma 2 are positive on the divisor A and negative on B . Because P, Q have opposite parity, one lies in A and the other in B . The residue relation gives

$$L := \sum_{T_i \in A} c_i \text{ev}_{T_i} = \sum_{T_i \in B} (-c_i) \text{ev}_{T_i}, \quad (2)$$

where all displayed coefficients are positive. Thus L has two $r = 3d$ atom representations, one containing P and the other Q .

We claim that L is in the interior of the projective moment cone $\mathcal{P}_{\overline{X}, 2d}^\vee$. If a nonzero section $s \in \mathcal{P}_{\overline{X}, 2d}$ satisfied $L(s) = 0$, positivity of the coefficients in the first representation of (2) would force s to vanish at every point of A . A nonnegative real section has even vanishing order at every real zero, so

$$\text{div}(s) \geq 2A.$$

Both sides have degree N , hence equality holds and $2A \sim \mathcal{L}$, contradicting (1). Therefore L is strictly positive on every nonzero element of the positivity cone, which is exactly interiority in the dual cone.

For a connected real elliptic curve, the source proves that the projective moment cone is generated using r evaluations. Its Lemma 6.2.3 further states that every interior functional has exactly two r -atomic representations. The two in (2) are consequently the only minimal projective representations of L . Both meet S , so no r -atomic representation is supported on the affine locus $X(\mathbb{R})$.

The degree- d moment matrix $M_d(L)$ is positive definite: for nonzero $g \in H^0(\overline{X}, \mathcal{O}(d))$, the square g^2 is a nonzero nonnegative section and hence $L(g^2) > 0$. Thus

$$\text{rank } M_d(L) = \dim H^0(\overline{X}, \mathcal{O}(d)) = 3d = r.$$

Every atomic representation of L therefore uses at least r atoms.

On the other hand, the source's Lemma 6.2.6 says that every interior projective moment functional has a representation using $r + 1$ atoms while avoiding any prescribed finite subset of $\overline{X}(\mathbb{R})$. Apply it to S . This proves that $L \in \mathcal{M}_{2d}(X(\mathbb{R}))$ and that

$$\mathcal{C}_{X, \leq 2d}(L) = r + 1, \quad \mathcal{C}_{\overline{X}, 2d}(L) = r.$$

A positive flat extension would, by the flat extension theorem, yield an r -atomic representing measure on $X(\mathbb{R})$, which we have just proved impossible. Therefore L has no positive flat extension. $\square$

Verification report

- **Degree count.** On a plane cubic, $\deg \mathcal{O}_{\overline{X}}(2d) = 6d$ and $h^0(\mathcal{O}_{\overline{X}}(2d)) = 6d$ by Riemann–Roch. The residue circuit therefore has exactly $3d$ positive and $3d$ negative terms.

- **Prescribed points.** The longer-arc construction makes the Abel sum map surjective and puts an even number of other zeros between P and Q , which is exactly the parity needed for opposite residue signs.
- **Interiority.** The only possible supporting nonnegative section would have divisor $2A$; the generic condition $2A \not\sim \mathcal{L}$ excludes it.
- **Minimal representations.** No new identifiability claim is assumed: exact two-representation uniqueness is precisely Lemma 6.2.3 of the source, applied after interiority is proved.
- **Affine membership.** Lemma 6.2.6 of the source supplies the required $(3d+1)$ -atom representation avoiding the entire finite infinity set.

No computation is used in the proof. The auxiliary script `code/numerical_circuit_check.py` verifies in degree $d = 1$ that six real intersections of a connected cubic and a real conic produce a three-positive/three-negative evaluation circuit and two equal positive-definite moment matrices.

Novelty and review recommendation

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2407.06017, the paper title, flat extensions, real points at infinity, affine Carathéodory numbers, and moment problems on plane cubics. No prior packet or attempt was found. A bounded external search using the exact conjecture sentence and its core phrases found the source paper but no later statement of a resolution. Because the current arXiv source is recent and the topic is specialized, novelty confidence is moderate rather than absolute.

Send for specialist review. The highest-value checks are: (i) the generic avoidance of $2A \sim \mathcal{L}$ inside the fixed-Abel-sum configuration, (ii) the sign comparison in the residue relation under projective evaluation normalizations, and (iii) that *pathological behavior* is interpreted as failure of ordinary positive flat extension, as in Example 7.2.3.

References

- [1] L. Baldi, G. Blekherman, and R. Sinn, *Nonnegative Polynomials and Moment Problems on Algebraic Curves*, arXiv:2407.06017, arxiv.org/abs/2407.06017.
- [2] K. Schmüdgen, *The Moment Problem*, Graduate Texts in Mathematics 277, Springer, 2017.